

Final Project — Data Science in Cyber

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Project Objective

The goal of this project is to critically evaluate a published article, blog post, or tutorial related to Data Science in Cybersecurity.

Students are expected not only to reproduce the proposed solution, but also to investigate whether the author's claims are supported by the data and experiments. A successful project demonstrates methodological rigor, critical thinking, and clear communication.

Source Selection

Select an article, blog post, or tutorial related to one of the following topics:

- Anomaly Detection
- Intrusion Detection Systems (IDS)
- Malware Detection
- Phishing Detection
- Fraud Detection
- Cybersecurity Graph Analytics
- Privacy and Data Protection
- Adversarial Machine Learning
- Time Series Analysis for Cybersecurity
- Any other cybersecurity topic approved by the lecturer

The selected source must:

- Clearly define a problem.
- Propose a solution.
- Include an implementation or a GitHub repository.
- Provide data or sufficient information to reproduce the experiment.

PDF Report Requirements

The report must be written in English and include the following sections.

1. Summary of the Source

Describe:

- The problem being addressed.
- Why the problem is important.
- The proposed solution.
- The dataset used.
- The model or methodology employed.

2. Critical Evaluation

Discuss:

- The main claims made by the author.
- Whether the claims are supported by the presented evidence.
- Whether the evaluation methodology is appropriate.
- Possible weaknesses or limitations.
- Whether the conclusions are justified.

If your findings contradict the author's conclusions, clearly explain why.

3. Feature Engineering Analysis

Analyze:

- Whether feature engineering was performed.
- Which features were used.
- Describe whether transformations, normalization, encoding, aggregation, feature creation, or feature selection were applied. Examples include logarithmic transformations, Box-Cox transformations, standardization, min-max scaling, one-hot encoding, target encoding, and dimensionality reduction techniques. For each transformation, explain why it was applied, what problem it was intended to address, and how it affected the analysis or model performance. Whenever possible, support your discussion with evidence showing whether the transformation improved data quality, model accuracy, interpretability, or overall predictive performance.
- Is there redundancy in the system? How would you spot it, and how would you tackle this issue?
- Whether the feature engineering process was meaningful. Provide an explanation that takes into account both the mathematical intuition and the cybersecurity aspect.
- Additional features that could potentially improve performance.

4. Reproducibility Analysis

Discuss:

- Whether the code can be executed successfully.
- Whether all required files and dependencies are available.
- Whether hidden preprocessing steps exist.
- The overall reproducibility of the work.

5. Experimental Results

Present:

- The experiments you performed.
- Any modifications you introduced.
- The models you trained.
- The evaluation metrics.
- The obtained results.

6. Conclusions

Summarize:

- Key findings.
- Lessons learned.
- Strengths and weaknesses of the proposed solution.
- Suggestions for future improvements.

Python Notebook Requirements

The notebook must be complete, executable, and clearly documented.

1. Data Loading

Include:

- Data loading.
- Data inspection.
- Data size, feature types.
- Temporal analysis.

- Missing value analysis.
- Analyze the columns and the index names. Does this make sense?
- Handling single values, features, or irrelevant
- duplicated features.

2. Exploratory Data Analysis (EDA)

The notebook must contain EDA, including:

- Feature distributions.
- Missing values analysis.
- Outlier analysis.
- If there are temporal features, ensure that the analysis takes this into account. Does this align with your world knowledge?
- Crosstab or group-by analysis. correlation analysis when appropriate and justify the methodology used. Discuss which correlation measure was selected and explain why it is suitable for the data and research question. Recall that we studied Pearson, Spearman, and Kendall correlation coefficients, each of which captures different types of relationships and relies on different assumptions.

For example:

- **Pearson correlation** is appropriate for measuring linear relationships between continuous variables.
- **Spearman correlation** is suitable for monotonic relationships and is more robust to outliers and non-normal data.
- **Kendall correlation** is often preferred for smaller datasets or when rank consistency is of primary interest.

Additional correlation measures may be used if they are better suited to the problem. For each method applied, explain its mathematical meaning, underlying assumptions, advantages, and limitations.

The report should also discuss why the selected method provides meaningful insight into the cybersecurity problem being investigated and whether the observed correlations are practically significant in addition to being statistically significant.

- Class imbalance analysis (prevalence). What is the real-world meaning of this? Do we have a sampling problem in one of the classes? Did the authors address this?
- Relevant visualizations.

3. Feature Engineering

Include relevant preprocessing, such as:

- Encoding categorical variables. Which method did you use and why?
- Feature scaling.
- Feature creation.
- Feature selection.
- Dimensionality reduction when appropriate.

4. Model Training

Train at least two models.

Examples include:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost
- Support Vector Machine
- K-Nearest Neighbors
- Isolation Forest
- Local Outlier Factor
- Autoencoders

5. Evaluation

Evaluation metrics are critical because they determine how success is measured, reveal a model's strengths and weaknesses, and ensure that its performance is assessed in a manner meaningful to the cybersecurity problem being addressed.

Metrics

Use appropriate evaluation metrics.

For classification:

- Accuracy
- Precision
- Recall
- F_1 Score
- F_β Score
- Matthews Correlation Coefficient (MCC)
- ROC-AUC
- Confusion Matrix

For anomaly detection:

- Precision
- Recall
- F_1 Score
- F_β Score
- ROC-AUC (when applicable)
- Matthews Correlation Coefficient (MCC)

For regression:

- MAE
- MSE
- RMSE
- R^2

Evaluation Metrics.

The evaluation section should include a detailed discussion of every metric used in the project. For each metric, provide both its mathematical definition and an interpretation within the cybersecurity context of the problem. Explain what the metric measures, why it is relevant, and how it reflects the practical performance of the system. For example, discuss the implications of False Positives and False Negatives in the specific application domain. For example, malware detection, intrusion detection, phishing detection, or fraud detection.

Students are encouraged to select evaluation metrics that are appropriate for their specific problem rather than using a fixed list blindly. Additional metrics may be introduced if they provide a more meaningful assessment of performance. Likewise, metrics that are not informative or suitable for the chosen task may be omitted. However, every such decision must be clearly documented and justified. The report should explain why particular metrics were selected, why others were excluded, and how the chosen metrics support the conclusions drawn from the analysis. A thoughtful and well-justified evaluation methodology will be valued more highly than the mechanical reporting of standard metrics.

6. Error Analysis

Discuss:

- Examples of model failures.
- Patterns in the errors.
- Cybersecurity implications of the errors.
- The trade-off between False Positives and False Negatives.

7. Executive Summary

Provide a concise summary of the entire project in approximately one page.

8. Summing It Up

The report must include a concise summary of the project, highlighting the problem, methodology, experiments, and key findings. Particular emphasis should be placed on the critical evaluation of the selected article, blog post, or tutorial. The objective is not to critique your own implementation, but rather to assess the strengths, weaknesses, assumptions, limitations, and validity of the original work. Your analysis should determine whether the author's claims are supported by evidence and whether the proposed methodology is appropriate for the problem being addressed.

The summary should include:

- The problem being addressed.
- The selected article, blog post, or tutorial.
- The dataset used.

- The methodology employed.
- The main findings of your reproduction study.
- Whether the author’s claims were supported by your results.
- The most important insights obtained from the analysis.
- Do you recommend using this project on similar problems
- The final conclusion of the project.

The purpose of this section is to allow a reader to quickly understand the project, its methodology, and its conclusions without reading the full report.

Code Quality Requirements

The code must follow good software engineering practices.

- Short and focused functions.
- Meaningful variable names.
- No unnecessary loops.
- Proper use of `pandas`, `numpy`, and `scikit-learn`.
- Clear separation between preprocessing, EDA, training, and evaluation.
- Accurate and informative comments written in English.
- Avoid duplicated code.
- Use fixed random seeds when appropriate.
- Use train/test split or cross-validation.

Submission Requirements

Submission must be performed using a public GitHub repository.

The repository must contain:

- PDF report.
- Python notebook.
- Any supporting code files.
- README file containing:
 - Project description.
 - Link to the selected article/blog/tutorial.
 - Link to the original GitHub repository.

- Execution instructions.
- Dataset source.

The GitHub repository link should be sent to the email address that will be provided by the examiner.

Submission Deadline

Friday, July 10, 2026, at 23:59.

Late submissions will not be accepted without prior approval.

Grading Rubric

Component	Points
Problem Understanding and Source Selection	10
Summary Quality	15
Critical Evaluation of the Author's Claims	20
Feature Engineering Analysis	10
Exploratory Data Analysis (EDA)	15
Model Training and Comparison	15
Evaluation and Error Analysis	10
Code Quality and Software Engineering	5
Total	100

Submission Guidelines

Each student must submit an individual project and is responsible for all submitted materials.

Students may discuss ideas and concepts with their peers; however, the report, code, analysis, and conclusions must represent the student's own work.

The course staff reserves the right to invite any student to a short oral discussion or demonstration of the project. During this meeting, the student may be asked to explain the methodology, implementation, experimental results, and conclusions presented in the submission.

Failure to demonstrate sufficient understanding of the submitted work may affect the project grade.

Important Note

The purpose of this project is not merely to reproduce results.

Students are expected to evaluate the methodology critically and objectively.

If the author's claims are not supported by the evidence, this should be clearly stated and justified.

A successful project demonstrates rigorous analysis, sound methodology, and thoughtful conclusions, regardless of whether the original claims are confirmed or rejected.